

LAKSHYA

JEE 2027

Lecture No.- 21

MATHEMATICS

RELATIONS & FUNCTIONS

By – Sachin Jakhar Sir



Topics *to be covered*



1 **Classification of Functions**

2 **Question Practice**

3

4



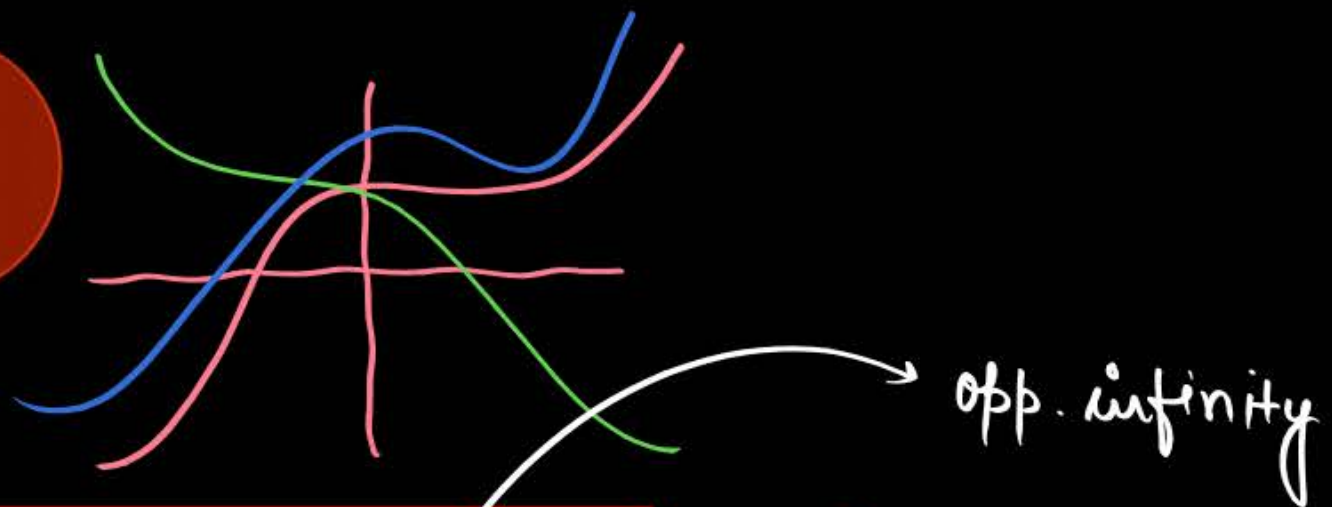


LAST CLASS REVISION



Answer the following Questions:

1. For $x \in R$, **Range of odd degree polynomial is R .** $\rightarrow \textcircled{T}$
2. For $x \in R$, **Range of even degree polynomial is proper subset of R .** $\rightarrow \textcircled{T}$
3. **Polynomial functions have continuous graphs.** $\rightarrow \textcircled{T}$
 (same infinity)
4. If $y \in [-3, 4]$ then $|y| \in \underline{[0, 4]}$.
5. If $y \in (-6, 2]$ then $|y| \in \underline{[0, 6]}$.
6. If $y \in [-2, 3)$ then $|2y - 3| \in \underline{[0, 7]}$.
7. $f(x) = \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}}$ & $g(x) = \tan x$ are **Identical functions.** $\rightarrow \textcircled{F}$
 $\sqrt{\tan^2 x} = |\tan x|$
8. $y = \tan x \cdot \cot x$ & $y = \sin x \cdot \operatorname{cosec} x$ are **Identical functions.** $\rightarrow \textcircled{F}$
 $\rightarrow x \neq n\pi/2$
 $x \neq n\pi$

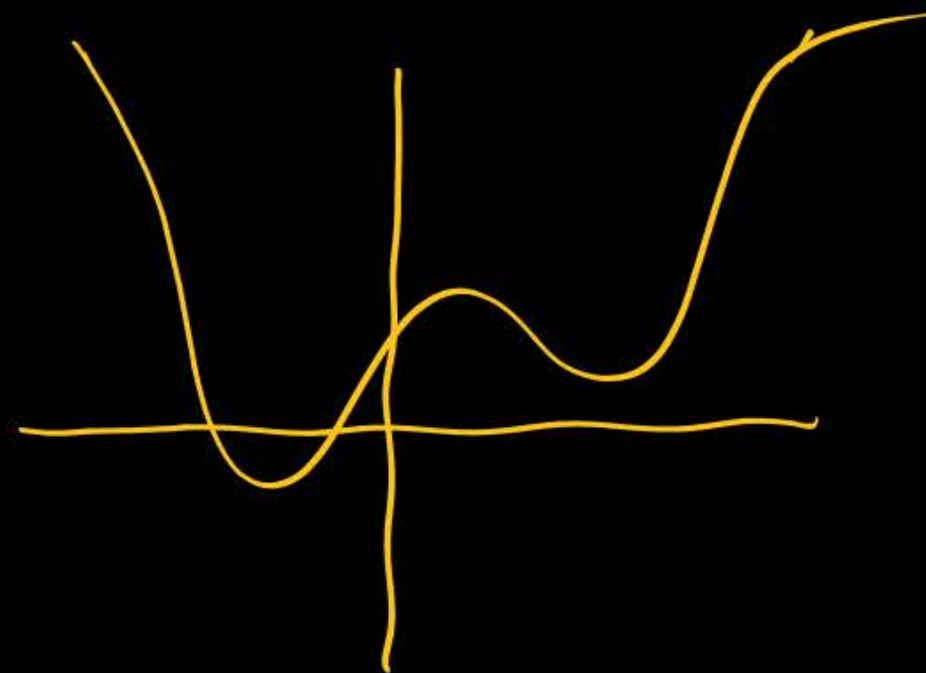


$$\# y = a\underline{x^3} + bx^2 + cx + d$$

$$\begin{array}{l} x \rightarrow \infty \Rightarrow (a > 0) \ y \rightarrow \infty \\ x \rightarrow -\infty \Rightarrow (a > 0) \ y \rightarrow -\infty \end{array} \quad (-\infty, \infty)$$

$$\# y = a\underline{x^4} + bx^3 + cx^2 + dx + e$$

$$\begin{array}{l} x \rightarrow \infty \rightarrow (a > 0) \ y \rightarrow \infty \\ x \rightarrow -\infty \rightarrow (a > 0) \ y \rightarrow \infty \end{array}$$





Question Practice

Challenger Question

6



Find the number of elements contained in the range of the function

$$f(x) = \left[\frac{x}{6} \right] \left[\frac{-6}{x} \right], \forall x \in (0, 30] \text{ (where } [.] \text{ denotes greatest integer function)}$$

Kab aur kitne?

$$\frac{x}{6} \in (0, 5] \Rightarrow \frac{6}{x} \in \left[\frac{1}{5}, \infty \right)$$

$$-\frac{6}{x} \in \left(-\infty, -\frac{1}{5} \right]$$

$$\frac{x}{6} = 1, 2, 3, 4, 5$$

$$x = 6, 12, 18, 24, 30$$

$$f(x) = \left[\frac{x}{6} \right] \left[\frac{-6}{x} \right] = 0 \times -1 = 0, x \in (0, 6)$$

$$= 1 \times -1 = -1, x \in (6, 12)$$

$$= 2 \times -1 = -2, x \in (12, 18)$$

$$\vdots$$

$$\left\{ \begin{array}{l} x=6 \rightarrow f(6) = 1 \times -1 = -1 \\ x=12 \rightarrow f(12) = 2 \times -1 = -2 \\ x=18 \rightarrow f(18) = -3 \\ x=24 \rightarrow f(24) = -4 \\ x=30 \rightarrow f(30) = -5 \end{array} \right.$$

$$\text{Range} \in \{0, -1, -2, -3, -4, -5\}$$

FUNCTIONS

CLASS-XII PORTION



CLASSIFICATION OF FUNCTIONS



100% Quesⁿ.
आसता

ONE-ONE & MANY-ONE FUNCTIONS

(I) One-One function (Injective Mapping) :

If every element of set A is mapped with different element of Set B.

(II) Many-One Function (Non-Injective Mapping) :

If two or more elements of set A have the same image in set B.

ONTO AND INTO FUNCTIONS

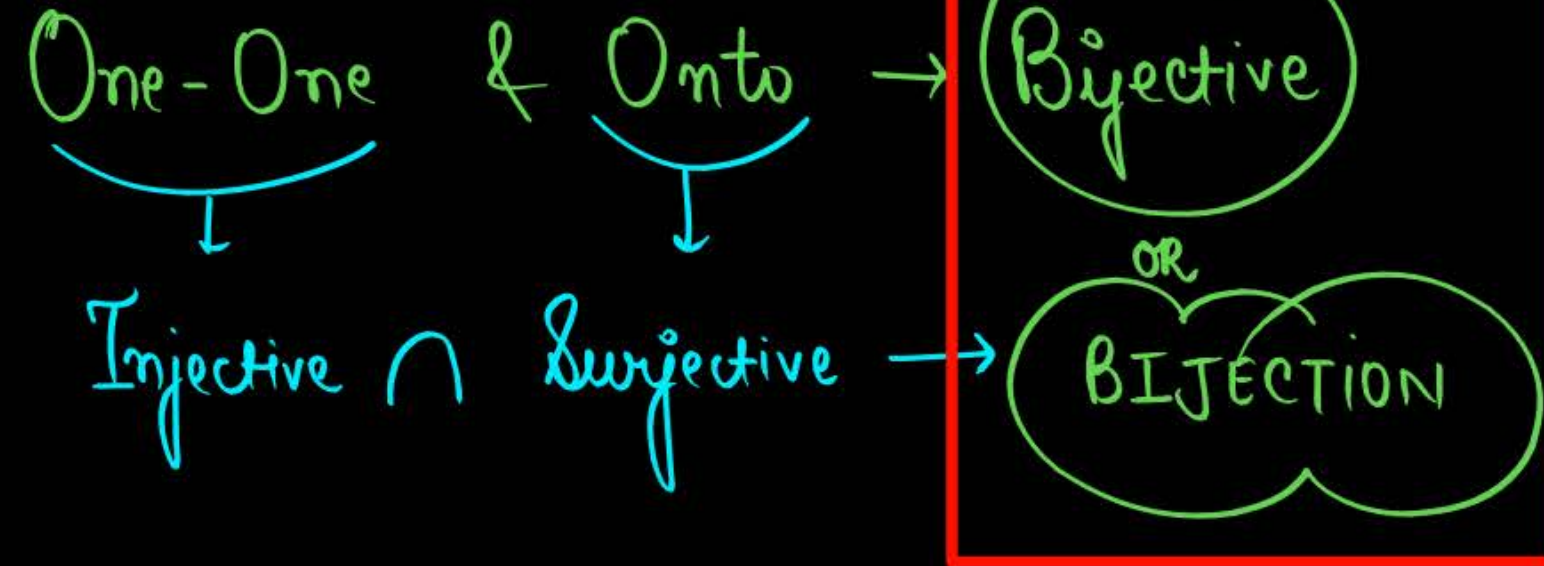
(I) Onto Function (Surjective Mapping) :

If each element in B (co-domain) is the image of at least one element in A, then we say that f is onto function.

(II) Into Function (Non-Surjective Mapping):

If there exists at least one element in co-domain which is not the image of any element in domain, then $f(x)$ is into.

BIJECTIVE MAPPING OR BIJECTIVE FUNCTIONS





HOW TO CHECK ONE-ONE OR MANY-ONE??

M-I: Graphical Method: Any line parallel to X-axis cuts the graph at 2 or more than 2 points, then function is many-one.

M-II: Using $\frac{dy}{dx}$:

If sign doesn't change $\rightarrow$ One-one.

$\frac{dy}{dx} \rightarrow \oplus / \ominus / + / -$

$\frac{dy}{dx} \rightarrow \oplus / \ominus$

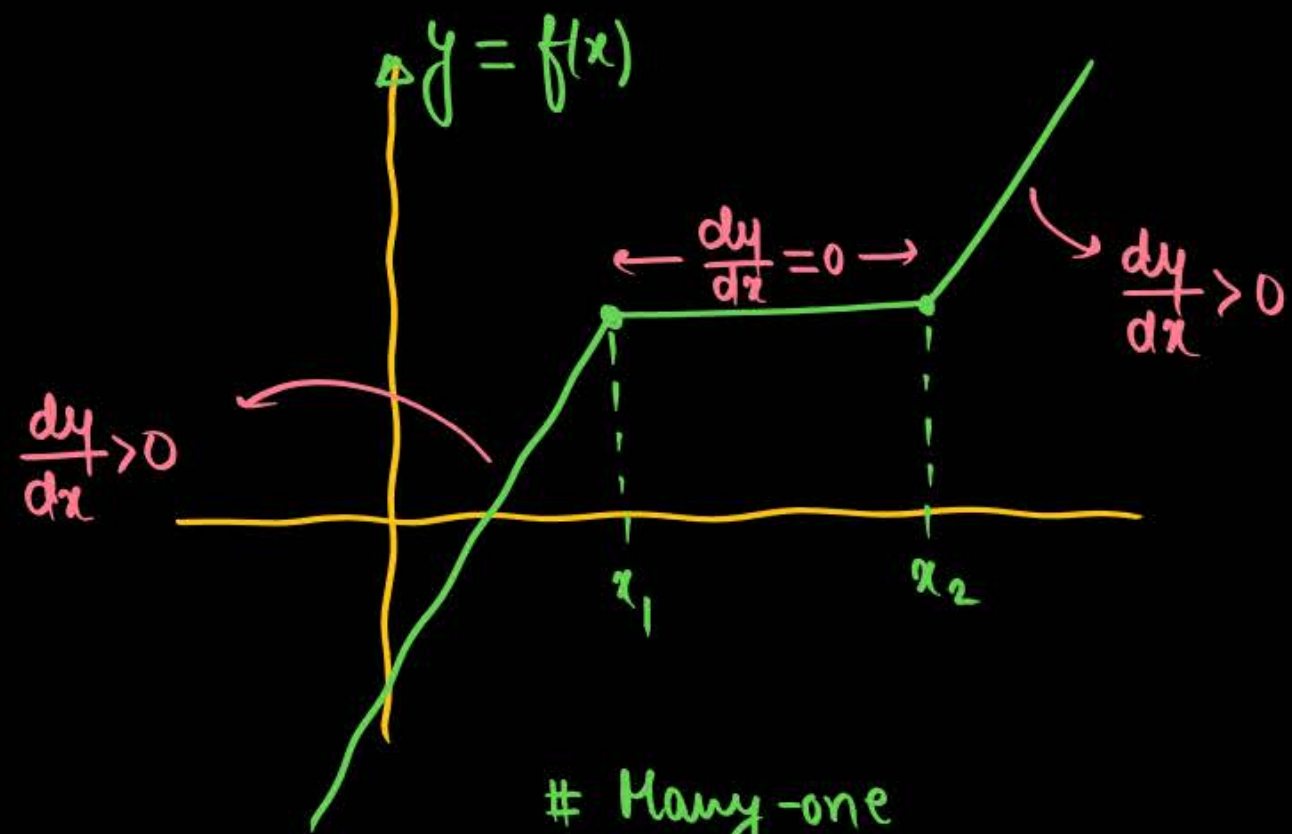
If sign changes $\rightarrow$ Many-One ✓

$\frac{dy}{dx} = 0 \rightarrow \text{sol}^n$.
set shouldn't form an interval.

M-III: By Observation: (Multiple roots) or (for different 'x' getting same y)

M-IV: Theoretical:

Solve $f(x_1) = f(x_2)$ & analyse. (Boards!)



Many-one

$$\# \frac{dy}{dx} = 0 \Rightarrow x \in (x_1, x_2)$$

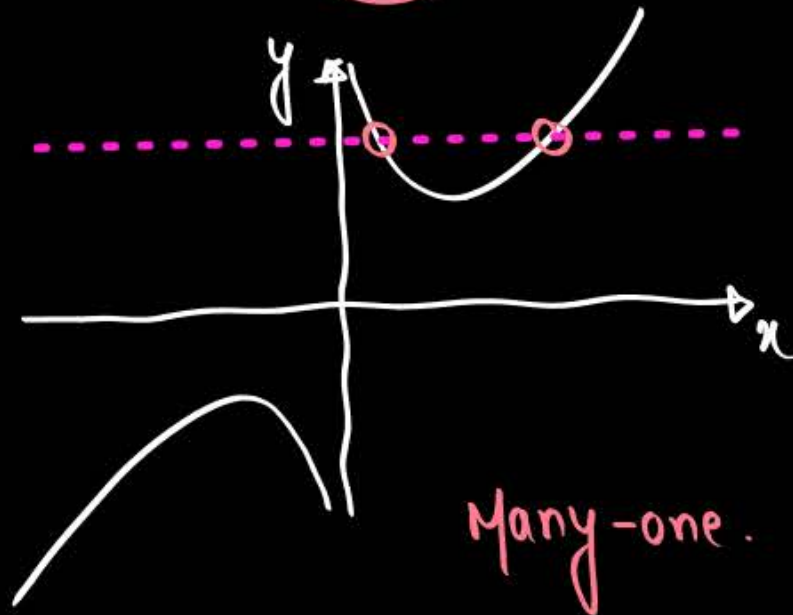
for ONE-ONE

$$\# \frac{dy}{dx} = 0$$

Solⁿ should not form an interval.

$f(x) = x + \frac{1}{x}$

M-1

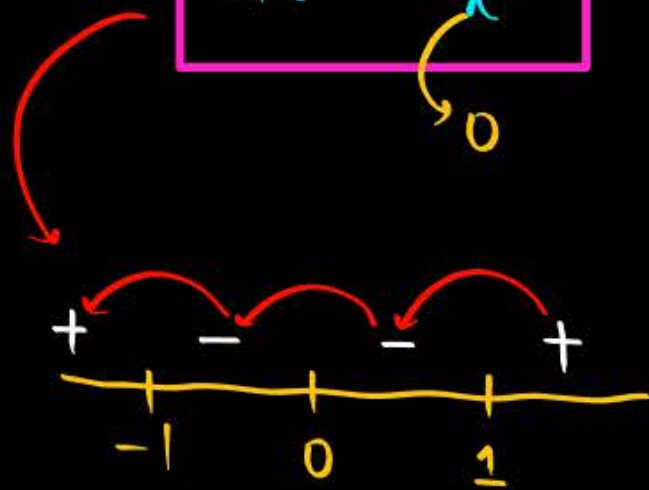


M-2

$f'(x) = \frac{dy}{dx} = 1 - \frac{1}{x^2}$

$\frac{dy}{dx} = \frac{x^2 - 1}{x^2} = \frac{(x+1)(x-1)}{x^2}$

Annotations: ± 1 points to the numerator, 0 points to the denominator.



Sign-change $\rightarrow$ Many-one.

$f(x) = x + \frac{1}{x}$ (Boards!)

H-4 :

egⁿ $f(x_1) = f(x_2)$

Solve.

only $x_1 = x_2 \rightarrow$ one-one.
or

$(x_1 = x_2)$ or other solⁿ $\rightarrow$ Many-one.

$$f(x_1) = f(x_2)$$

$$x_1 + \frac{1}{x_1} = x_2 + \frac{1}{x_2}$$

$$x_1 - x_2 + \frac{1}{x_1} - \frac{1}{x_2} = 0$$

$$(x_1 - x_2) + \left(\frac{x_2 - x_1}{x_1 x_2} \right) = 0$$

$$(x_1 - x_2) \left\{ \frac{1}{x_1} - \frac{1}{x_2} \right\} = 0$$

$$(x_1 - x_2)(x_1 x_2 - 1) = 0$$

$\left. \begin{array}{l} x_1 = x_2 \\ x_1 x_2 = 1 \end{array} \right\} \text{Many-one.}$

example

$$\# x_1 x_2 = 1$$

$$x_1 = 2, x_2 = \frac{1}{2}$$

$$f(2) = f(x_1) = 2 + \frac{1}{2} = \frac{5}{2}$$

$$f\left(\frac{1}{2}\right) = f(x_2) = \frac{1}{2} + 2 = \frac{5}{2}$$

Acidic Question

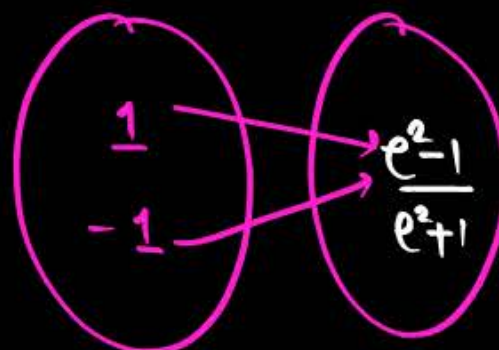
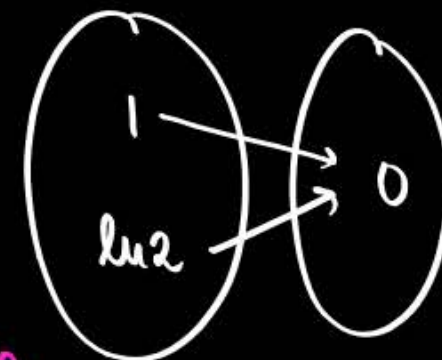


Check the following functions as one-one or many-one:

(i) $y = (e^x - 2)(\sin^2 x - 5\sin x + 6)(\ln x)$

(ii) $f(x) = \frac{e^{x^2} - e^{-x^2}}{e^{x^2} + e^{-x^2}}$

(iii) $f(x) = x + \cos x$



Many-one.



$f(x) = x + \cos x$

One-One / Injective

$f'(x) = \frac{dy}{dx} = 1 - \sin x$

$\oplus$

$\square$

$+ \quad 0 \quad + \quad 0 \quad + \quad 0 \quad + \quad 0 \quad +$

$-\frac{3\pi}{2} \quad \frac{\pi}{2} \quad \frac{5\pi}{2} \quad \frac{9\pi}{2} \quad \dots$

$\frac{dy}{dx} = 0$ ka solⁿ set is not forming an interval

One-One
Many-One



4 Method

Onto
Into



Calculate
Range.

Check following functions as **ONTO/INTO**?

(i) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x)$ is an odd degree polynomial. $\rightarrow$ Onto

(ii) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x)$ is an even degree polynomial. $\rightarrow$ Into

(iii) $f: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1], f(x) = \sin x \rightarrow$ Onto

★★ (iv) $f: \mathbb{R} \rightarrow \underline{\mathbb{R}}, f(x) = x^3 + x^2 + 2x + \sin x \rightarrow$ Onto

★★ (v) $f: \mathbb{R} \rightarrow \underline{\mathbb{R}}, f(x) = \frac{e^{-x} - e^{-|x|}}{e^x + e^{-x}} \in [0, 1)$

$\left. \begin{array}{l} x \rightarrow \infty \Rightarrow y \rightarrow \infty \\ x \rightarrow -\infty \Rightarrow y \rightarrow -\infty \end{array} \right\} \text{Range } (-\infty, \infty) \rightarrow \mathbb{R}$

$= \frac{e^{-x} - e^{-x}}{e^x + e^{-x}} = 0, x \geq 0$

$= \frac{e^{-x} - e^x}{e^x + e^{-x}} \in (0, 1), x < 0$

$\rightarrow$ Into

$$\# \quad y = \frac{e^{-x} - e^x}{e^x + e^{-x}}, \quad x < 0$$

$$x \in (-\infty, 0)$$

$$2x \in (-\infty, 0)$$

$$\# \quad e^{2x} \in (0, 1)$$

$$y = \frac{2}{e^{2x} + 1} - 1$$

$$e^{2x} \in (0, 1)$$

$$e^{2x} + 1 \in (1, 2)$$

$$\frac{2}{e^{2x} + 1} \in (1, 2)$$

$$y = \frac{\frac{1}{e^x} - e^x}{e^x + \frac{1}{e^x}}$$

$$y = \frac{1 - e^{2x}}{e^{2x} + 1}$$

$$y e^{2x} + y = 1 - e^{2x}$$

$$e^{2x}(y+1) = 1-y$$

$$(0, 1)$$

$$e^{2x} = \frac{1-y}{y+1}$$

$$0 < \frac{1-y}{y+1} < 1$$

Question [JEE Mains-2023]



Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = \frac{x^2+2x+1}{x^2+1}$. Then

☒ A $f(x)$ is many-one in $(-\infty, -1)$

☒ B $f(x)$ is many-one in $(1, \infty)$

☒ C $f(x)$ is one-one in $[1, \infty)$ but not in $(-\infty, \infty)$

☒ D $f(x)$ is one-one in $(-\infty, \infty)$

Many-one.

$$f(x) = \frac{x^2+2x+1}{x^2+1}$$

$$\# \frac{dy}{dx} = \frac{(2x+2)(x^2+1) - (x^2+2x+1)2x}{(x^2+1)^2}$$

$$\frac{dy}{dx} = \frac{\cancel{2x^3} + 2x^2 + \cancel{2x} + 2 - \cancel{2x^3} - 4x^2 - \cancel{2x}}{(x^2+1)^2}$$

$$\frac{dy}{dx} = \frac{2 - 2x^2}{(x^2+1)^2} = \frac{2(1-x)(1+x)}{(x^2+1)^2}$$

$$\begin{array}{c} - & 0 & + & 0 & - \\ \hline & -1 & & 1 & \end{array}$$

~~A~~ Injective only

1-1

1-1 | Many-one ✓

Onto

1-1 | Many-one ✓ ↘ Onto

Diagram illustrating the mapping of elements from one set to another:

- Left Set: 2 and $\frac{2}{3}$
- Right Set: $\frac{1}{2}$
- Mappings:
 - $2 \rightarrow \frac{1}{2}$
 - $\frac{2}{3} \rightarrow \frac{1}{2}$

$x \in (0, \infty)$
 $\frac{1}{x} \in (0, \infty)$
 $-\frac{1}{x} \in (-\infty, 0)$
 $1 - \frac{1}{x} \in (-\infty, 1)$

$\left| 1 - \frac{1}{x} \right| \in \underbrace{[0, \infty)}_{\text{Range}}$

$$1 - \frac{1}{x} \in (-\infty, 1)$$

Question [JEE Mains-2019]



Let a function $f : (0, \infty) \rightarrow [0, \infty)$ be defined by $f(x) = \left| 1 - \frac{1}{x} \right|$. Then f is

- ☒ **A** Injective only
 $\leftarrow 1-1$
- ☒ **B** Both injective as well as surjective
 $\leftarrow$ Onto
- ☒ **C** Not injective but it is surjective
 $\leftarrow$ Many-one
- ☐ **D** Neither injective nor surjective

Many-one

$$y = 1 - \frac{1}{x}, \quad x \geq 1$$

$$= \frac{1}{x} - 1, \quad 0 < x < 1$$

$$\frac{dy}{dx} = \left(\frac{1}{x^2} \right), \quad +ve$$

$$= \left(-\frac{1}{x^2} \right), \quad -ve$$

DO IT BY YOURSELF (DIBY)

CLASSIFICATION OF FUNCTIONS

DIBY-141

Classify following function defined as:

(i) $f(x) = x^2 + \ln x$

(ii) $f(x) = 9^{|x|} + 9^{2-x}$

[Ans. : (i) $f(x)$ is bijective,

(ii) $f(x)$ is many-one and into function]

DIBY-142

$f: [0, 3] \rightarrow A$ such that $f(x) = x^2 - 2x + 5$. Find A if $f(x)$ is onto?

[Ans. : $A = [4, 8]$]

DIBY-143

Classify following function as injective or surjective:

(i) $f: [-1, 3] \rightarrow [-8, 72]$ be defined as $f(x) = 4x^3 - 12x$,

(ii) $f: \mathbb{R} \rightarrow \{x \in \mathbb{R} : -1 < x < 1\}$, $f(x) = \frac{x}{1+|x|}$

(iii) $f: [-1, 3] \rightarrow [-37, 27]$ $f(x) = 2x^3 - 6x^2 - 18x + 17$

(iv) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^{x^2} + \cos x$

(v) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x - \sin x$,

Ans. :

(i) many-one and onto

(ii) one-one and onto

(iii) one-one and onto

(iv) many-one and into

(v) one-one and onto

CLASSIFICATION OF FUNCTIONS

DIBY-144

The function $f(x) = \frac{x^2+2x-15}{x^2-4x+9}, x \in R$

[JEE Mains-2024]

- (A) both one-one and onto. (B) onto but not one-one.
(C) neither one-one nor onto. (D) one-one but not onto.

[Ans.: (C)]

DIBY-145

The function $f: N - \{1\} \rightarrow N$; defined by $f(n)$ = the highest prime factor of n , is:

[JEE Mains-2024]

- (A) Both one-one and onto (B) One-one only
(C) Onto only (D) Neither one-one nor onto

[Ans.: (D)]

DIBY-146

Let $f: R \rightarrow R$ be a function such that $f(x) = \frac{x^2+2x+1}{x^2+1}$. Then

[JEE Mains-2023]

- (A) $f(x)$ is many-one in $(-\infty, -1)$ (B) $f(x)$ is many-one in $(1, \infty)$
(C) $f(x)$ is one-one in $[1, \infty)$ but not in $(-\infty, \infty)$ (D) $f(x)$ is one-one in $(-\infty, \infty)$

[Ans.: (C)]

CLASSIFICATION OF FUNCTIONS

DIBY-147

The function $f: [0, 3] \rightarrow [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is: **[JEE-2012]**

- | | |
|--------------------------|------------------------------|
| (A) one-one and onto | (B) onto but not one-one |
| (D) one-one but not onto | (D) neither one-one nor onto |

[Ans.: (B)]

DIBY-148

Let a function $f: N \rightarrow N$ be defined by $f(n) = \begin{cases} 2n, & n = 2, 4, 6, 8, \dots \\ n-1, & n = 3, 7, 11, 15, \dots \\ \frac{n+1}{2}, & n = 1, 5, 9, 13, \dots \end{cases}$ then, f is

[JEE Mains-2022]

- | | |
|------------------------------|--------------------------|
| (A) One-one but not onto | (B) Onto but not one-one |
| (C) Neither one-one nor onto | (D) One-one and onto |

[Ans.: (D)]

CLASSIFICATION OF FUNCTIONS

DIBY-149

Let N be the set of natural numbers and two functions f and g be defined as

$f, g: N \rightarrow N$ such that $f(n) = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ is odd} \\ \frac{n}{2} & \text{if } n \text{ is even} \end{cases}$ and $g(n) = n - (-1)^n$. Then $f \circ g$ is:

[JEE Mains-2019]

- | | |
|----------------------------|-------------------------------|
| (A) Onto but not one-one. | (B) One-one but not onto. |
| (C) Both one-one and onto. | (D) Neither one-one nor onto. |

[Ans. : (A)]

DIBY-150

Let $A = \{x \in R: x \text{ is not a positive integer}\}$. Define a function $f: A \rightarrow R$ as $f(x) = \frac{2x}{x-1}$
then f is

[JEE Mains-2019]

- | | |
|----------------------------------|--------------------------------------|
| (A) Injective but not surjective | (B) Not injective |
| (C) Surjective but not injective | (D) Neither injective nor surjective |

[Ans. : (A)]

CLASSIFICATION OF FUNCTIONS

DIBY-151

Let $f: R \rightarrow R$ be defined as $f(x) = 2x - 1$ and $g: R - \{1\} \rightarrow R - \{1\}$ be defined as

$g(x) = \frac{x - \frac{1}{2}}{x - 1}$. Then the composition function $f(g(x))$ is [JEE Mains-2021]

- | | |
|------------------------------|--------------------------|
| (A) Both one-one and onto | (B) Onto but not one-one |
| (C) Neither one-one nor onto | (D) One-one but not onto |

[Ans.: (D)]

DIBY-152

Let $f: R \rightarrow R$ defined as $f(x) = \frac{2x^2 - 3x + 2}{3x^2 + x + 3}$. Then f is [JEE Mains-2026]

- | | |
|---------------------------|------------------------------|
| (A) Both one-one and onto | (B) One-one but not onto |
| (C) Onto but not one-one | (D) neither one-one nor onto |

[Ans.: (D)]

DIBY-153

The function $f: (-\infty, \infty) \rightarrow (-\infty, 1)$, defined by $f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$ is **[JEE Mains-2025]**

- | | |
|--------------------------|------------------------------|
| (A) Onto but not one-one | (B) Both one-one and onto |
| (C) One-one but not onto | (D) Neither one-one nor onto |

[Ans. : (c)]



Homework



Re-attempt all good Questions of Class

DIBY

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians